Measuring Causes and Consequences of Errors in RNA

Jakub Koubele

TAC meeting - Progress Report
2nd March, 2026

Introduction

My PhD project consists of two main parts:

Introduction

My PhD project consists of two main parts:

1. Statistical model for estimating changes in transcription elongation speed from total RNA-seq data

Introduction

My PhD project consists of two main parts:

1. Statistical model for estimating changes in transcription elongation speed from total RNA-seq data
2. Calling of mismatches in transcript from single-cell RNA-seq data, and integration of it with single-cell data analysis.

Introduction

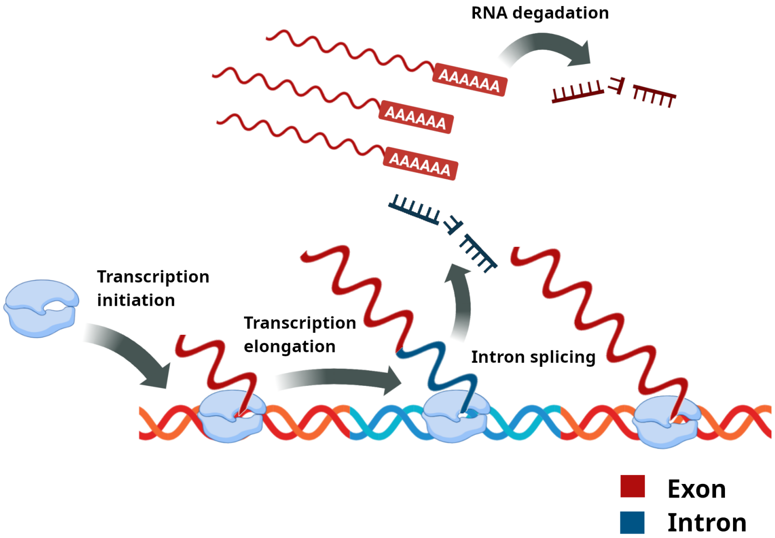
My PhD project consists of two main parts:

1. Statistical model for estimating changes in transcription elongation speed from total RNA-seq data
2. Calling of mismatches in transcript from single-cell RNA-seq data, and integration of it with single-cell data analysis.

The first part turned into a more complex modeling of the RNA metabolism, and I focused almost entirely on it during the last year.

RNA metabolism overview

RNA metabolism overview



RNA metabolism as a dynamical system

RNA metabolism as a dynamical system

For a given gene, following molecules are present in the system:

RNA metabolism as a dynamical system

For a given gene, following molecules are present in the system:

- Nascent and mature transcripts: T_{nascent}, T_{mature}.

RNA metabolism as a dynamical system

For a given gene, following molecules are present in the system:

- Nascent and mature transcripts: $T_{\text{nascent}},\ T_{\text{mature}}$.
- Nascent and finished (but unspliced) introns: $I_{\text{nascent}}, I_{\text{unspliced}}$.

RNA metabolism as a dynamical system

For a given gene, following molecules are present in the system:

- Nascent and mature transcripts: T_{nascent}, T_{mature}.
- Nascent and finished (but unspliced) introns: $I_{\mathrm{nascent}}, I_{\mathrm{unspliced}}$.

The dynamics may be described by an ODE system:

RNA metabolism as a dynamical system

For a given gene, following molecules are present in the system:

- Nascent and mature transcripts: T_{nascent}, T_{mature}.
- Nascent and finished (but unspliced) introns: $I_{\text{nascent}}, I_{\text{unspliced}}$.

The dynamics may be described by an ODE system:

$$\begin{aligned}
\dot{T}_{\text{nascent}} &= k_{\text{init}} - k_{\text{elong}} \cdot T_{\text{nascent}} \cdot L_{\text{gene}}^{-1} \\
\dot{I}_{\text{nascent}} &= k_{\text{init}} - k_{\text{elong}} \cdot I_{\text{nascent}} \cdot L_{\text{intron}}^{-1} \\
\dot{T}_{\text{mature}} &= k_{\text{elong}} \cdot T_{\text{nascent}} \cdot L_{\text{gene}}^{-1} - k_{\text{deg}} \cdot T_{\text{mature}} \\
\dot{I}_{\text{unspliced}} &= k_{\text{elong}} \cdot I_{\text{nascent}} \cdot L_{\text{intron}}^{-1} - k_{\text{splice}} \cdot I_{\text{unspliced}}
\end{aligned}$$

where we have

- Kinetic rates: $[k_{\text{init}}] = \text{polymerases} \cdot \text{s}^{-1}$, $[k_{\text{elong}}] = \text{bp} \cdot \text{s}^{-1}$, $[k_{\text{splice}}] = \text{s}^{-1}$, $[k_{\text{deg}}] = \text{s}^{-1}$
- Gene and intron lenghts: $[L_{\text{gene}}] = [L_{\text{intron}}] = \text{bp}$

Steady state solution

Steady state solution

We may solve the ODE system for steady state solution:

$$T_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{gene}}$$

$$I_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{intron}}$$

$$T_{\text{mature}} = \frac{k_{\text{init}}}{k_{\text{deg}}}$$

$$I_{\text{unspliced}} = \frac{k_{\text{init}}}{k_{\text{splice}}}$$

Steady state solution

We may solve the ODE system for steady state solution:

$$T_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{gene}}$$

$$I_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{intron}}$$

$$T_{\text{mature}} = \frac{k_{\text{init}}}{k_{\text{deg}}}$$

$$I_{\text{unspliced}} = \frac{k_{\text{init}}}{k_{\text{splice}}}$$

- The steady state quantities depends only on the ratios of kinetic rates.

Steady state solution

We may solve the ODE system for steady state solution:

$$T_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{gene}}$$

$$I_{\text{nascent}} = \frac{k_{\text{init}}}{k_{\text{elong}}} \cdot L_{\text{intron}}$$

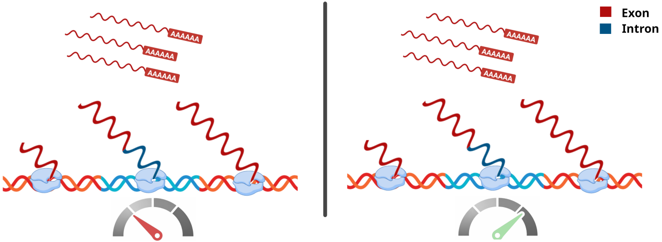
$$T_{\text{mature}} = \frac{k_{\text{init}}}{k_{\text{deg}}}$$

$$I_{\text{unspliced}} = \frac{k_{\text{init}}}{k_{\text{splice}}}$$

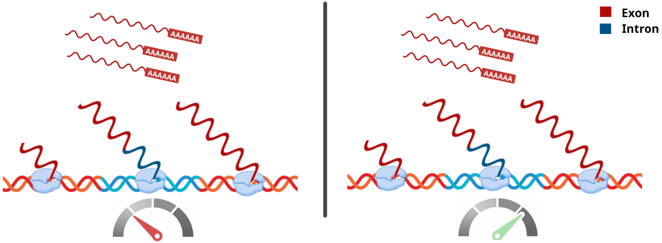
- The steady state quantities depends only on the ratios of kinetic rates.
- Therefore, scaling all kinetic rate by some constant $c \in \mathbb{R}_+$ does not change the steady state.

Non-identifiability of kinetic rates from steady state

Non-identifiability of kinetic rates from steady state

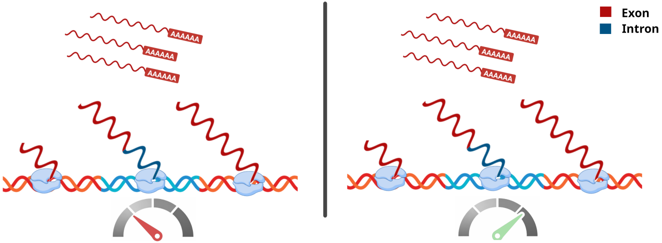


Non-identifiability of kinetic rates from steady state



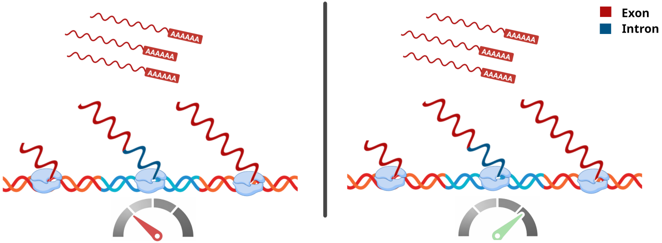
- Only ratios of kinetic rates, e.g., $\frac{k_{\mathrm{elong}}}{k_{\mathrm{deg}}}$, are identifiable.

Non-identifiability of kinetic rates from steady state



- Only ratios of kinetic rates, e.g., $\frac{k_{\text{elong}}}{k_{\text{deg}}}$, are identifiable.
- When comparing samples (treatment vs. control), only differences of LFC can be estimated and statistically tested.

Non-identifiability of kinetic rates from steady state



- Only ratios of kinetic rates, e.g., $\frac{k_{\text{elong}}}{k_{\text{deg}}}$, are identifiable.
- When comparing samples (treatment vs. control), only differences of LFC can be estimated and statistically tested.
- E.g., we may estimate the value of

$$\mathrm{LFC}(k_{\text{elong}}) - \mathrm{LFC}(k_{\text{deg}})$$

and test the hypothesis

$$\mathrm{H}_0 : \quad \mathrm{LFC}(k_{\text{elong}}) - \mathrm{LFC}(k_{\text{deg}}) = 0$$

Non-identifiability is a common problem

Non-identifiability is a common problem

- Consider an autonomous dynamical system with state vector $\boldsymbol{x}$, and with dynamics
$$\dot{\boldsymbol{x}} = f(\boldsymbol{x})$$

Non-identifiability is a common problem

- Consider an autonomous dynamical system with state vector $\boldsymbol{x}$, and with dynamics
$$\dot{\boldsymbol{x}} = f(\boldsymbol{x})$$

- It holds that
$$f(\boldsymbol{x}) = 0 \Leftrightarrow c \cdot f(\boldsymbol{x}) = 0 \quad \forall c \in \mathbb{R}_+$$

Non-identifiability is a common problem

- Consider an autonomous dynamical system with state vector $\boldsymbol{x}$, and with dynamics

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x})$$

- It holds that

$$f(\boldsymbol{x}) = 0 \Leftrightarrow c \cdot f(\boldsymbol{x}) = 0 \quad \forall c \in \mathbb{R}_+$$

- Therefore, steady-state data cannot distinguish $f(\boldsymbol{x})$ from $c \cdot f(\boldsymbol{x})$. Consequently, only ratios of rates are identifiable.

Non-identifiability is a common problem

- Consider an autonomous dynamical system with state vector $\boldsymbol{x}$, and with dynamics

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x})$$

- It holds that

$$f(\boldsymbol{x}) = 0 \Leftrightarrow c \cdot f(\boldsymbol{x}) = 0 \quad \forall c \in \mathbb{R}_+$$

- Therefore, steady-state data cannot distinguish $f(\boldsymbol{x})$ from $c \cdot f(\boldsymbol{x})$. Consequently, only ratios of rates are identifiable.
- This type of structural non-identifiability is common in RNA metabolism models.

Dealing with structural non-identifiability

Dealing with structural non-identifiability

- Use orthogonal experiment to measure at least one kinetic rate independently.

Dealing with structural non-identifiability

- Use orthogonal experiment to measure at least one kinetic rate independently.
- Perturb the biological system to obtain non-steady-state observations.

Dealing with structural non-identifiability

- Use orthogonal experiment to measure at least one kinetic rate independently.
- Perturb the biological system to obtain non-steady-state observations.
    - E.g., inhibiting/releasing transcription, and/or using labeled nucleotides.

Compositional non-identifiability in RNA-seq

Compositional non-identifiability in RNA-seq

- RNA-seq data are compositional: we observe proportions within samples, not absolute abundances. This introduces an additional source of non-identifiability.

Compositional non-identifiability in RNA-seq

- RNA-seq data are compositional: we observe proportions within samples, not absolute abundances. This introduces an additional source of non-identifiability.
- Let

$$S_i = \{\text{RNA molecules in sample } i\}$$

  We are typically interested in some subset $A_i \subset S_i$, and we aim to compare its size between samples, i.e. $\frac{|A_2|}{|A_1|}$.

Compositional non-identifiability in RNA-seq

- RNA-seq data are compositional: we observe proportions within samples, not absolute abundances. This introduces an additional source of non-identifiability.
- Let

$$S_i = \{\text{RNA molecules in sample } i\}$$

  We are typically interested in some subset $A_i \subset S_i$, and we aim to compare its size between samples, i.e. $\frac{|A_2|}{|A_1|}$.
- With RNA-seq, we observe the number of reads r_i corresponding to the molecules of interest, which scales as

$$r_i \propto \frac{|A_i|}{|S_i|} = \frac{|A_i|}{|A_i| + |A_i^c|}$$

  where A_i^c is a complement of A_i in S_i.

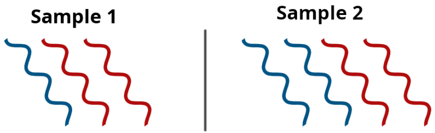
Compositional non-identifiability: example

Consider the following example of comparing intronic reads (blue) between two samples:

Compositional non-identifiability: example

Consider the following example of comparing intronic reads (blue) between two samples:

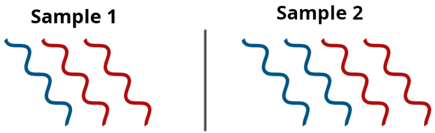
- If sample 2 has slower intron splicing, it will contain more intronic reads:



Compositional non-identifiability: example

Consider the following example of comparing intronic reads (blue) between two samples:

- If sample 2 has slower intron splicing, it will contain more intronic reads:



- If sample 2 has faster RNA degradation, it will contain fewer exonic reads:

Compositional non-identifiability: example

Consider the following example of comparing intronic reads (blue) between two samples:

- If sample 2 has slower intron splicing, it will contain more intronic reads:

- If sample 2 has faster RNA degradation, it will contain fewer exonic reads:

- In both cases, the fraction of intronic reads is identical, making the two scenarios indistinguishable from RNA-seq data.

Conclusion & open questions

Conclusion & open questions

- Estimating changes in RNA kinetic rates from total RNA-seq data poses two non-identifiability problems:

Conclusion & open questions

- Estimating changes in RNA kinetic rates from total RNA-seq data poses two non-identifiability problems:
  - Structural non-identifiability: the underlying dynamical system cannot be identified from steady-state data.

Conclusion & open questions

- Estimating changes in RNA kinetic rates from total RNA-seq data poses two non-identifiability problems:
    - Structural non-identifiability: the underlying dynamical system cannot be identified from steady-state data.
    - Compositional non-identifiability: arises because RNA-seq measures only relative abundances within a sample.

Conclusion & open questions

- Estimating changes in RNA kinetic rates from total RNA-seq data poses two non-identifiability problems:
  - Structural non-identifiability: the underlying dynamical system cannot be identified from steady-state data.
  - Compositional non-identifiability: arises because RNA-seq measures only relative abundances within a sample.
- To overcome these issues, we need either additional measurements, and/or different types of data than total RNA-seq.

Conclusion & open questions

- Estimating changes in RNA kinetic rates from total RNA-seq data poses two non-identifiability problems:
  - Structural non-identifiability: the underlying dynamical system cannot be identified from steady-state data.
  - Compositional non-identifiability: arises because RNA-seq measures only relative abundances within a sample.
- To overcome these issues, we need either additional measurements, and/or different types of data than total RNA-seq.

The End

Thank you for your attention!

Figures were created using BioRender.com.